System Design Document (SDD)

SEMKO Integrated Management System

System Name: SEMKO Enterprise Resource Planning System

Developer: In-House Development Team

Lead Developer: Dancan kariuki

Version: 1.0

Date: March 2026

Document Type: Technical Design Specification (For Director Approval)

Document Control

Version	Date	Author	Changes
1.0	March 2026	In-House Team	Initial draft for director approval

1. Executive Summary

1.1 Purpose

This System Design Document (SDD) provides the technical blueprint for building SEMKO's integrated management system. As the lead in-house developer, I am presenting this document to the director for approval before development begins. This document outlines WHAT we will build, HOW we will build it, and WHEN it will be delivered.

1.2 System Overview

SEMKO is a fleet management company offering:

- Transport Services: Hauling materials from quarries to construction sites
- Material Supply: Purchasing and reselling materials (tarmac, ballast, dust, etc.) from quarries
- Fleet Operations: Managing owned and hired trucks
- Maintenance: In-house workshop for truck repairs

1.3 Core Objectives

Objective	Description	Business Impact
Full Automation	Eliminate manual paperwork and calculations	Reduce errors, save time
Configurability	All business rules are changeable via UI—NO hardcoding	Adapt to regulations instantly
Real-Time Visibility	Operations, payroll, and inventory at your fingertips	Better decision-making
Cloud-Based	Accessible anywhere, no on-premise server	Lower IT costs, work from anywhere
Audit Trail	Every change tracked	Compliance, dispute resolution

---

2. System Architecture

2.1 High-Level Architecture

```text
┌──────────────────────────────────────────────────────────────┐
│                      CLIENT LAYER                            │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐  ┌──────────┐    │
│  │ Desktop  │  │ Tablet   │  │ Mobile   │  │ API      │    │
│  │ Browser  │  │ (Stores) │  │ (Field)  │  │ Clients  │    │
│  └──────────┘  └──────────┘  └──────────┘  └──────────┘    │
└──────────────────────────────────────────────────────────────┘
                              │
                              ▼
┌──────────────────────────────────────────────────────────────┐
│                      APPLICATION LAYER                       │
│  ┌──────────────────────────────────────────────────────┐   │
│  │              Django (Python) Backend                  │   │
│  │  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌────────┐ │   │
│  │  │ Trip     │ │ Payroll  │ │ Stores   │ │ Vehicle│ │   │
│  │  │ Module   │ │ Module   │ │ Module   │ │ Module │ │   │
│  │  └──────────┘ └──────────┘ └──────────┘ └────────┘ │   │
│  │  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌────────┐ │   │
│  │  │ Client   │ │ Cess     │ │ Fuel     │ │ Mainte-│ │   │
│  │  │ Module   │ │ Module   │ │ Module   │ │ nance  │ │   │
│  │  └──────────┘ └──────────┘ └──────────┘ └────────┘ │   │
│  │  ┌──────────────────────────────────────────────┐  │   │
│  │  │         Rules Engine (Fully Configurable)     │  │   │
│  │  └──────────────────────────────────────────────┘  │   │
│  └──────────────────────────────────────────────────────┘   │
└──────────────────────────────────────────────────────────────┘
                              │
                              ▼
┌──────────────────────────────────────────────────────────────┐
│                      DATA LAYER                              │
│  ┌──────────────────────────────────────────────────────┐   │
│  │              PostgreSQL Database                      │   │
│  │  • Relational Data Structure                          │   │
│  │  • JSON Fields for Flexible Config                    │   │
│  │  • Audit Triggers                                     │   │
```

```
│   └──────────────────────────────────────────────────────┘   │
│   ┌──────────────────────────────────────────────────────┐   │
│   │              Cloud File Storage                        │   │
│   │   • Delivery Note Scans                                │   │
│   │   • Cess Receipt Scans                                 │   │
│   │   • Driver Documents                                   │   │
│   └──────────────────────────────────────────────────────┘   │
└──────────────────────────────────────────────────────────────┘
```

2.2 Technology Stack (Approved for Development)

Layer	Technology	Justification
Backend API	Python Django + Django REST Framework	Robust, secure, rapid development, excellent ORM
Database	PostgreSQL	Reliable, JSON support, performance
Frontend	React with TypeScript (TSX)	Modern, maintainable, type-safe
Styling	Tailwind CSS	Rapid UI development, consistent design
Hosting	Cloud VPS (DigitalOcean/AWS/Linode)	No on-premise server, accessible anywhere
File Storage	Cloud Storage (S3-compatible)	Secure document storage
Backup	Automated daily + off-site	Disaster recovery

2.3 Why This Stack?

- Django: Built-in admin interface, authentication, and ORM speed up development
- React + TypeScript: Type safety prevents bugs, better developer experience
- Tailwind: No custom CSS, faster UI development
- Cloud: No server room needed, work from anywhere, automatic backups

---

3. Core Principle: FULL Configurability

3.1 The Rules Engine Philosophy

NO HARDCODING BUSINESS RULES. Everything that can change must be configurable through the admin UI. This means:

- When government changes tax rates → Update in UI, not code
- When bonus structure changes → Update in UI, not code
- When cess rates change → Update in UI, not code
- When new trip types needed → Add in UI, not code

3.2 What We Can Configure (Without Developer)

Category	Configurable Items	Update Frequency
Statutory	PAYE bands & rates, NSSF tiers, SHIF %, Housing Levy %	As gazetted
Bonus Rules	Trip conditions, bonus amounts per vehicle type	Business changes

Trip Types	"Far Location" keywords, special trip definitions	Route changes
Cess	Rates by material + county, effective dates	Monthly/negotiated
Vehicle	Daily rates per type, maintenance schedules	As needed
Materials	Material types, default prices	Market changes
Stores	Reorder levels, min/max stock	As needed

3.3 How It Works (Simple Explanation)

Instead of writing code like:

```python
if trip_count == 3 and vehicle_type == "Tata":
    bonus = 800
```

We build a system where you can do this in the UI:

1. Go to Admin → Bonus Rules
2. Click "Add Rule"
3. Set: Vehicle Type = Tata, Trip Count = 3 → Bonus = 800
4. Save

Next time rules change: You change it in the UI, not me changing code.

---

4. Core Modules Overview

4.1 Trip Management Module (Heart of the System)

Complete Weighbridge Workflow (Automated)

```text
QUARRY SITE                                CLIENT SITE
═══════════════════════════════════════════════════════════════════════

STEP 1: EMPTY           ──Load──▶   STEP 3: LOADED
Weighbridge #1                       Weighbridge #3
(Empty Weight)                        (Gross Weight at Site)
        │                                    │
        ▼                                    ▼
STEP 2: LOADED          ◀──Unload───   STEP 4: EMPTY
Weighbridge #2                       Weighbridge #4
(Gross Weight at Quarry)               (Empty Weight)

        │                                    │
        └──────────────┬──────────────────────┘
                       ▼
             DISCREPANCY CHECK
         Quarry Net = W2 - W1
         Site Net   = W3 - W4
         Discrepancy = Quarry Net - Site Net

         if discrepancy > threshold:
             PENALTY = discrepancy × (material_cost + transport_cost)
```

What Gets Automated:

- Weight calculations (no manual math)
- Discrepancy detection (alerts on shortage)
- Penalty calculation (auto-deduct from driver)
- Cess calculation (based on configurable rates)
- Trip classification (far/near/tarmac/dust(material) for bonuses)

4.2 Payroll Module

Complete Payroll Automation

```text
                  GROSS PAY
                       │
        ┌──────────────┼──────────────┐
        ▼              ▼              ▼
   Daily Rate       Bonuses        Allowances
   (Days × Rate)   (Trip based)     (If any)
        │              │              │
        └──────────────┼──────────────┘
                       ▼
                TOTAL GROSS PAY
                       │
        ┌──────────────┼──────────────┐
        ▼              ▼              ▼
     PAYE            SHIF          Housing Levy
   (Tax Table)     (2.75% now)      (1.5% each)
        │              │              │
        └──────────────┼──────────────┘
                       ▼
             TOTAL DEDUCTIONS
                       │
                       ▼
                  NET PAY
                       │
        ┌──────────────┼──────────────┐
        ▼                             ▼

   Bank Transfer                   Cash Payment
```

Statutory Deductions (All Configurable)

Deduction	Who Pays	Current Rate	Configurable?
PAYE (Tax)	Employee	Tax bands	YES - bands & rates
NSSF Tier I	Both	Fixed amounts	YES - tiers
NSSF Tier II	Both	Fixed amounts	YES - tiers
SHIF	Employee	2.75%	YES - percentage
Housing Levy	Both	1.5% each	YES - both rates
NITA	Employer	KES 50	YES - amount

4.3 Stores/Inventory Module

Complete Inventory Tracking

```text
┌──────────────┐
│  SUPPLIER    │
└──────────────┘
       │ Purchase Order
       ▼

┌──────────────┐      ┌──────────────────┐
│ GOODS        │──────►│ STOCK RECEIVING  │
│ RECEIVED     │      │ • GRN Created    │
└──────────────┘      │ • Stock Added    │
                      └──────────────────┘
                               │
                               ▼

                      ┌──────────────────┐
                      │   STOCK TABLE    │
                      │ • Current Qty    │
                      │ • Min/Max levels│
                      │ • Location       │
                      └──────────────────┘
                               │
                               │
              ┌────────────────┼────────────────┐
              ▼                ▼                ▼

┌──────────────────┐ ┌──────────────────┐ ┌──────────────────┐
│ WORKSHOP ISSUE   │ │ DRIVER ISSUE     │ │ OFFICE ISSUE     │
│ • To Mechanic    │ │ • To Driver      │ │ • To Staff       │
│ • For Vehicle    │ │ • For Trip       │ │ • For Office     │
│ • Requisition #  │ │ • Chargeable?    │ │ • Requisition #  │

└──────────────────┘ └──────────────────┘ └──────────────────┘
```

4.4 Cess Management (County Tax Automation)

Problem: Cess rates change by county, material, and maybe negotiations/agreements. Currently manual.

Solution: Configurable cess matrix in the system:

Material	County	Rate Type	Rate	Effective Period
Tarmac	Nairobi	Per Trip	500	2026-2027
Ballast	Nairobi	Per Trip	300	2026-2027
Tarmac	Kiambu	Per Trip	450	2026-2027

When trip entered: System auto-calculates cess based on current rules.

4.5 Hired Truck Handling

Problem: Hired truck payments are separate from driver payroll.

Solution: System tracks:

- Hired trucks separately
- Owner payment calculations
- Separate settlement reports

---

5. User Interface (React + TypeScript + Tailwind)

5.1 Key Screens (Mockups Ready)

Delivery Note Entry Screen (Mirrors Paper Form)

- Tab-based navigation: Quarry → Cess → Site → Summary
- Auto-calculation as you type
- Document scanner/upload
- Quick copy from previous trips

Dashboard (Real-Time Overview)

- Today's trips counter
- Pending deliveries
- Low stock alerts
- Driver performance
- Quick links to common tasks

Store Requisition (Simple 3-Step)

1. Select vehicle/mechanic
2. Search/add parts (with compatibility filter)
3. Submit for approval

5.2 Mobile Responsive

- Desktop: Full functionality (office)
- Tablet: Optimized for stores (large buttons)
- Phone: Field access for drivers (view trips, payslips, request parts)

---

6. Development Phases & Timeline

Phase 1: Foundation (Weeks 1) - COMPLETE BY APRIL 2026

- ✅ Development environment setup
- ✅ Database design
- ✅ User authentication & roles
- ✅ Basic UI framework
- ✅ Audit trail foundation

Phase 2: Master Data (Weeks 2-4) - COMPLETE BY APRIL 2026

- Vehicle management
- Driver management
- Client management (Corporate & Individual)
- Quarry management
- Material management
- Import existing data

Phase 3: Core Operations - Trips (Weeks 5-8) - COMPLETE BY MAY 2026

- Delivery Note entry form
- Weighbridge data capture
- Cess calculation engine (configurable)
- Penalty calculation
- Document upload
- Trip listing and search
- Hired truck handling

Phase 4: Rules Engine (Weeks 9-12) - COMPLETE BY JUNE 2026

- Configurable rules framework
- Bonus rules admin UI

- Cess rules admin UI
- Trip classification admin UI
- Statutory rates admin UI

Phase 5: Payroll Module (Weeks 13-16) - COMPLETE BY JULY 2026

- Daily rate calculation
- Bonus aggregation engine
- PAYE tax engine (configurable)
- NSSF engine (configurable)
- SHIF calculation
- Housing levy calculation
- NITA handling
- Payroll processing workflow
- Payslip generation

Phase 6: Stores Module (Weeks 17-20) - COMPLETE BY AUGUST 2026

- Item master
- Stock receiving
- Stock issuance
- Requisition workflow
- Stock adjustments
- Inventory reports

Phase 7: Fuel & Maintenance (Weeks 21-24) - COMPLETE BY SEPTEMBER 2026

- Fuel transaction entry
- Fuel consumption tracking
- Maintenance records
- Service scheduling alerts

Phase 8: Reporting

- All standard reports (30+ reports)
- Dashboard widgets
- Export functionality (PDF, Excel)
- Scheduled email reports

Phase 9: Testing & Deployment

- Unit testing
- Integration testing
- User acceptance testing
- Cloud server setup
- Data migration
- GO-LIVE

Phase 10: Training & Documentation

- User manual
- Admin guide
- Video tutorials
- On-site training sessions

6.1 Alternative Phased Rollout (Best option)

For faster business value, we can release modules as they're ready:

Release	Modules Included	Business Value
Release 1	Trip Management + Cess	Stop manual trip errors, automate cess
Release 2	Payroll + Statutory	Stop payroll calculation errors
Release 3	Stores Module	Real-time inventory control
Release 4	Fuel + Maintenance	Track all vehicle costs
Release 5	Full Reporting	Complete business visibility

Benefit: You get value earlier, and we can adjust based on feedback.

---

7. Cost Estimation (In-House Development)

7.1 Development Cost (Internal)

Resource	Time Allocation	Cost (Internal)
Lead Developer (You)	7.5 months full-time	Existing salary
Operations Manager (Input)	Part-time	Existing
Finance (Input)	Part-time	Existing
Testers (Staff)	2 weeks UAT	Existing

Total Direct Cost: Existing staff time reallocated

7.2 External/Hosting Costs (Monthly)

Item	Estimated Monthly Cost
Cloud VPS	$20-40 (KES 2,500-5,000)
Cloud Storage	$5-10 (KES 650-1,300)
Domain/SSL	$1-2 (KES 130-260)
Backup Storage	$5 (KES 650)
TOTAL MONTHLY	$31-57 (KES 4,000-7,500)

7.3 One-Time Setup Costs

Item	Cost
Domain registration (if new)	$15 (KES 2,000)
SSL Certificate (free with Let's Encrypt)	$0
TOTAL SETUP	$15 (KES 2,000)

7.4 Comparison: Off-the-Shelf Alternatives

Option	Initial Cost	Monthly	Customization	Data Ownership
In-House Development	Staff time	KES 5,000	Complete	100% Ours
Off-the-Shelf Fleet System	KES 500,000+	KES 20,000+	Limited	Vendor controlled
Manual/Excel	KES 0	Staff time	Flexible	100% Ours but error-prone

Verdict: In-house development is most cost-effective long-term with full control.

---

8. Risk Assessment & Mitigation

Risk	Probability	Impact	Mitigation
Scope creep (adding features mid-development)	High	High	Strict change control process; new features go to Phase 11
Data loss during migration	Low	High	Test migration with copy first; full backups
User resistance (staff prefer old ways)	Medium	Medium	Involve users in testing; good training; show time savings
Regulation changes during dev	Medium	Medium	Configurable rules mean changes are UI updates, not code rewrites
System downtime	Low	Medium	Cloud VPS with 99.9% uptime SLA; monitoring alerts
Developer leaving	Low	High	Document everything; code in company repository; knowledge transfer

---

9. Success Metrics (How We Measure)

9.1 Technical KPIs

Metric	Target
Page load time	< 3 seconds
Report generation	< 10 seconds
System uptime	99.5%
Data accuracy	100% (no calculation errors)

9.2 Business KPIs

Metric	Target
Payroll processing time	2 hours
Trip data entry errors	< unknown to 0%
Inventory accuracy	> 98%
Cess payment reconciliation	Real-time
Driver query resolution	Instant (self-service)

10. Approval Request

1. Approve this design document ✓ / ✗
2. Confirm development start date _____
3. Choose rollout approach:
    - Full release after 5.5 months
    - Phased releases (value earlier)
4. Assign key personnel for input:
    - Operations contact: _____
    - Finance contact: _____
    - Stores contact: _____
5. Budget approval for monthly cloud costs (KES ~5,000/month)

Next Steps After Approval:

1. Set up development environment
2. Create test database
3. Begin Phase 1 (Foundation)
4. Weekly/monthly progress reports to you

---

11. Summary

What We're Building

A complete, cloud-based system that automates:

- ✅ Trip management with weighbridge and cess
- ✅ Payroll with all statutory deductions (configurable)
- ✅ Stores/inventory control
- ✅ Fuel tracking
- ✅ Maintenance scheduling
- ✅ 30+ reports for full visibility
- And more as will be requested

Key Benefits

- No more manual calculations (errors eliminated)
- Configurable rules (you change rates, not code)
- Real-time visibility (know your business instantly)
- Cloud-based (access anywhere, no server room)
- Complete audit trail (every change tracked)

Investment

- Time: <5.5 months development
- Monthly cost: ~KES 5,000 (cloud hosting)
- People: Existing staff reallocated
- Continuous available maintenance

Return

- Payroll time: unknown → 2 hours
- Errors: unknown% → near zero
- Inventory: Unknown → real-time accurate
- Compliance: Manual → automated

Document Prepared By:

_____
Dancan
SEMKO In-House Team
Date: March 2026

Approval Signatures:

Role	Name	Signature	Date
Director	_____	_____	_____
Operations Manager	_____	_____	_____
Finance Manager	_____	_____	_____

This document represents the complete technical design for SEMKO's integrated management system. I am confident this system will transform our operations and provide excellent return on our development investment.